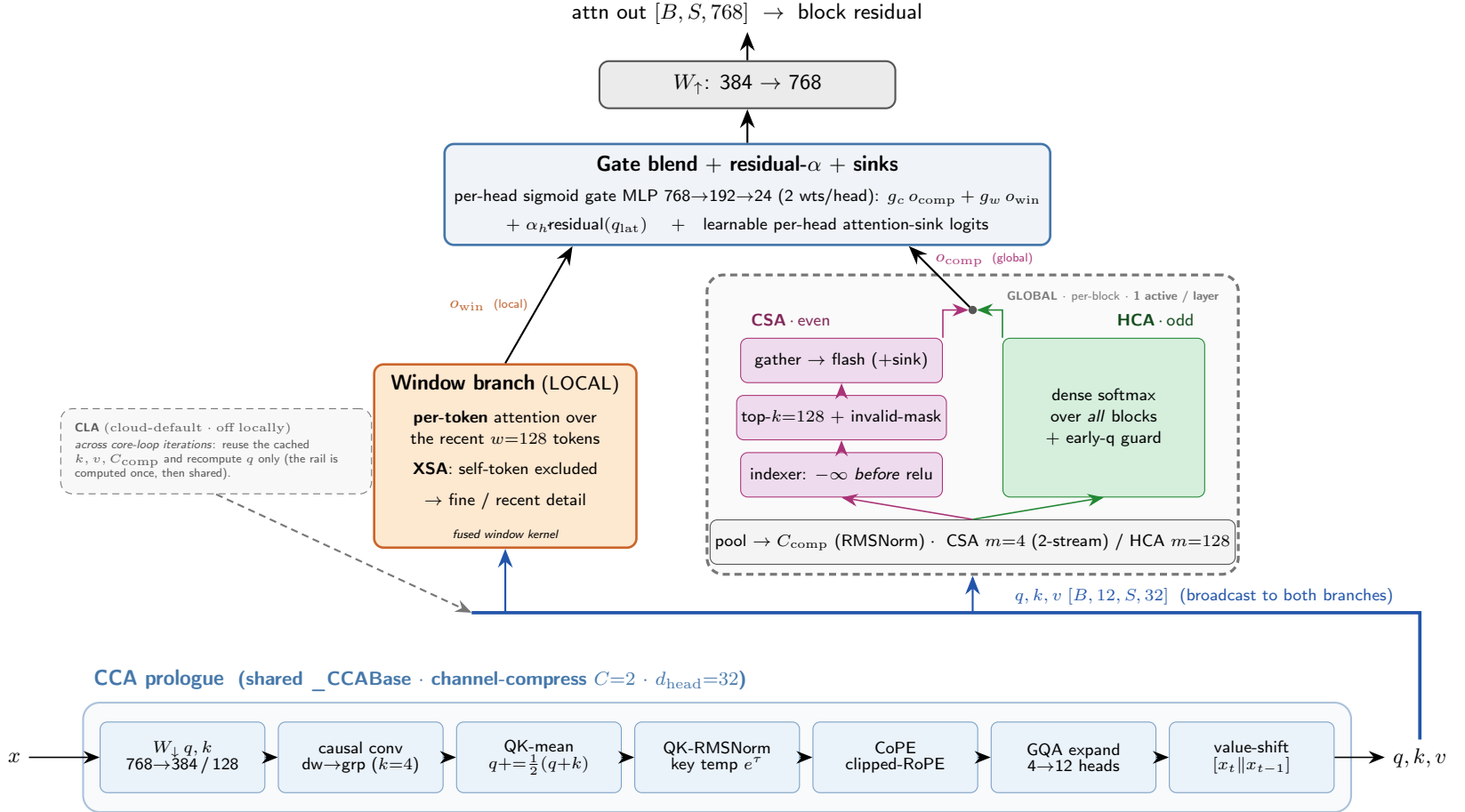


MORPH Attention — Triple-Axis Compression

a shared CCA channel-compress front-end feeds a local window || one global-compressed branch (CSA / HCA), blended by a per-head gate



Four compression axes (faithful to morph/model/attention.py): channel CCA $E \rightarrow E/2$ select CSA top-128 of $S/4$ blocks dense HCA all $S/128$ blocks local window $w=128$.

CSA/HCA alternate on the *global* layer index (even → CSA, odd → HCA), baked at `__init__` — a looped core layer keeps its variant on every iteration (*not* iteration-conditioned). Causal-leak fix: indexer masks future blocks to $-\infty$ before relu, and re-checks validity after gather.

Combine: per-head sigmoid gate → 2 weights (compressed, window); residual- α adds the down-projected q_{lat} back; attention sinks = a learnable per-head logit on the compressed softmax. Fused Triton: CCA prologue, conv, window, CSA, HCA (each fwd/grad-exact). **Notation:** $H=12$ q-heads, $H_{kv}=4$ kv (GQA $\times 3$), $C=2$, $d_{\text{head}}=32$, $n_{\text{blk}}=S/m$. Color secondary — every box is text-labeled.